

Draft Paper Outline (arXiv-style)

Title

- From 3D LiDAR to 2D Segmentation and Back: A Practical Pipeline for Building Footprints and Multiclass Point-Cloud Labeling

Authors

- Saviour Formosa — PhD Professor (Associate), University of Malta
- Tram Nguyen — PhD student, University of Malta
- Dylan Seychell — PhD, Faculty Member, University of Malta
- Alexey Kozhakin — PhD student, University of Malta
- TODO: confirm author order
- TODO: add emails/affiliations formatting

Abstract

- We present a practical end-to-end pipeline for LiDAR processing that converts 3D point clouds into 2D feature maps for segmentation and projects predictions back to 3D for GIS-ready outputs.
- The method uses KNN-based 3D→2D feature encoding and U-Net (ResNet-34) models for both binary building segmentation and 20-class labeling, followed by tile stitching and polygon extraction.
- We train on STPLS3D and demonstrate inference on Malta scenes (St Paul's Bay, Xewkija, Gozo Rabat, Sliema), producing building footprints and classified LAS files.
- The pipeline achieves METRICS_TBD on STPLS3D and yields high-quality qualitative results on Malta, enabling efficient building contour generation and 3D visualization.
- TODO: insert mIoU/IoU/F1 values for STPLS3D (binary + multiclass) and report inference/processing time per tile (and hardware).

1. Introduction

- Motivation: large LiDAR datasets are heavy to process; manual workflows are slow; GIS-ready outputs are needed.
- Problem statement: automate building extraction and multiclass labeling from raw LAS with minimal manual steps.
- Contributions (draft):
 - End-to-end pipeline from LAS to building polygons and GIS-ready outputs.
 - 3D→2D feature encoding with KNN on a regular grid for efficient 2D segmentation.
 - Dual models: binary building segmentation and 20-class segmentation.
 - 3D back-projection to LAS with class + RGB for visualization and downstream use.
- TODO: refine contribution wording + novelty claim

2. Related Work

- 3D point cloud segmentation (direct 3D networks, voxel/point approaches).
- 2D projection-based LiDAR processing for efficiency.
- Building footprint extraction and GIS polygonization.

- TODO: write this section as 1–3 short paragraphs (not bullets); compare approaches and position our pipeline.
- TODO: add 6–10 sources (papers + datasets).
- TODO: add citation to the “3D Data Science” book (PDF) and summarize current approaches from it.
- TODO: identify comparison methods and cite them.

3. Method

3.1 Overview

- Pipeline: tiling → feature encoding → 2D segmentation → stitching → polygons → 3D back-projection.
- Two applications:
 - Polygon generation app for multiple LAS files.
 - 3D multiclass segmentation app for a single LAS file.

3.2 Data Preprocessing and Tiling

- LAS input (LAS 1.0–1.4) with XYZ, RGB, class if present.
- Tile size: 250m × 250m.
- Point limit: 30,000 points per tile; if fewer, resample with replacement.
- Coordinate normalization: subtract min; scale to [0, 1] for 2D mapping.
- Malta inference naming: filename parsed as `x_km_y_km`, used to place 250m tiles within 1km blocks.
- TODO: exact dataset sizes and split details

3.3 3D-to-2D Feature Encoding

- Grid size: 512 × 512.
- KNN: K=4 via cKDTree.
- Features:
 - `z_mean` with 2D smoothing and profile correction.
 - `n_z` and `n_r` from local std of XYZ (surface normal proxy).
 - RGB via mean over K neighbors.
 - class channel via neighbor class (optional).
- TODO: justify feature set and smoothing choice

3.4 2D Segmentation Models

- U-Net with ResNet34 encoder, `in_channels=3`.
- Binary building segmentation: 2 classes.
- Multiclass segmentation: 20 classes with fixed color map.
- Inference batch size in code: 8.
- Loss: cross-entropy (per report).
- Optimizer: Adam (per report).
- TODO: training hyperparameters (epochs, LR, scheduler), augmentation

3.5 Post-processing and Polygon Generation

- Tile stitching by coordinates from filename; missing tiles filled with black.
- Binary thresholding on predicted mask.

- Contour detection (OpenCV), external contours.
- Minimum area filtering: min_area=500.
- Export polygons to Shapefile (SHP + DBF + SHX).
- **TODO: polygon accuracy evaluation (if any)**

3.6 3D Back-Projection

- Nearest-neighbor interpolation from 2D mask to 3D points (NearestNDInterpolator).
- Writes classification + RGB back to LAS; adds RGB dimensions if missing.

4. Experiments

4.1 Datasets

- STPLS3D used for training/validation (urban LiDAR).
- Malta inference areas: St Paul's Bay, Xewkija, Gozo Rabat, Sliema.
- **TODO: exact STPLS3D split, number of tiles/scenes, class distribution**
- **TODO: Malta dataset size, coordinate system, availability of ground truth**

4.2 Training Protocol

- Optimizer: Adam (per report).
- Loss: cross-entropy (per report).
- **TODO: epochs, batch size, LR, scheduler, augmentations, hardware**

4.3 Metrics

- Metrics computed in training notebook: precision, recall, IoU, F1.
- **TODO: report values (overall + per-class)**

5. Results

5.1 Quantitative Results

- **TODO: table with STPLS3D metrics for binary and multiclass models**

5.2 Qualitative Results

- Feature visualizations: elevation/surface normals/RGB (Gozo Rabat, Xewkija, St Paul's Bay).
- Building segmentation outputs (same Malta areas).
- Multiclass segmentation outputs (same Malta areas).
- 3D colored/classified LAS (e.g., Sliema).
- **TODO: pick final figures and captions for paper**

6. Applications

- GIS building footprints (Shapefile export).
- Urban planning / infrastructure mapping.
- Environmental monitoring via multiclass outputs.
- **TODO: discuss additional applications enabled by the method.**

7. Limitations

- 3D→2D projection may lose vertical structure.
- Sensitivity to point density and tiling boundaries.
- Generalization to unseen sensors/regions not fully tested.
- TODO: check "3D Data Science" book for relevant limitations to cite or discuss.

8. Conclusion and Future Work

- TODO: summary of results and practical impact.
- TODO: future work (better 3D supervision, larger datasets, more classes, ablations).

9. Acknowledgements

- TODO: funding, collaborators, dataset providers (STPLS3D, Malta sources, Glados sumpercomputer)

10. References

- TODO: bib entries for segmentation, LiDAR datasets, polygon extraction